

# Servo Arm Control System

## 1. Product Overview

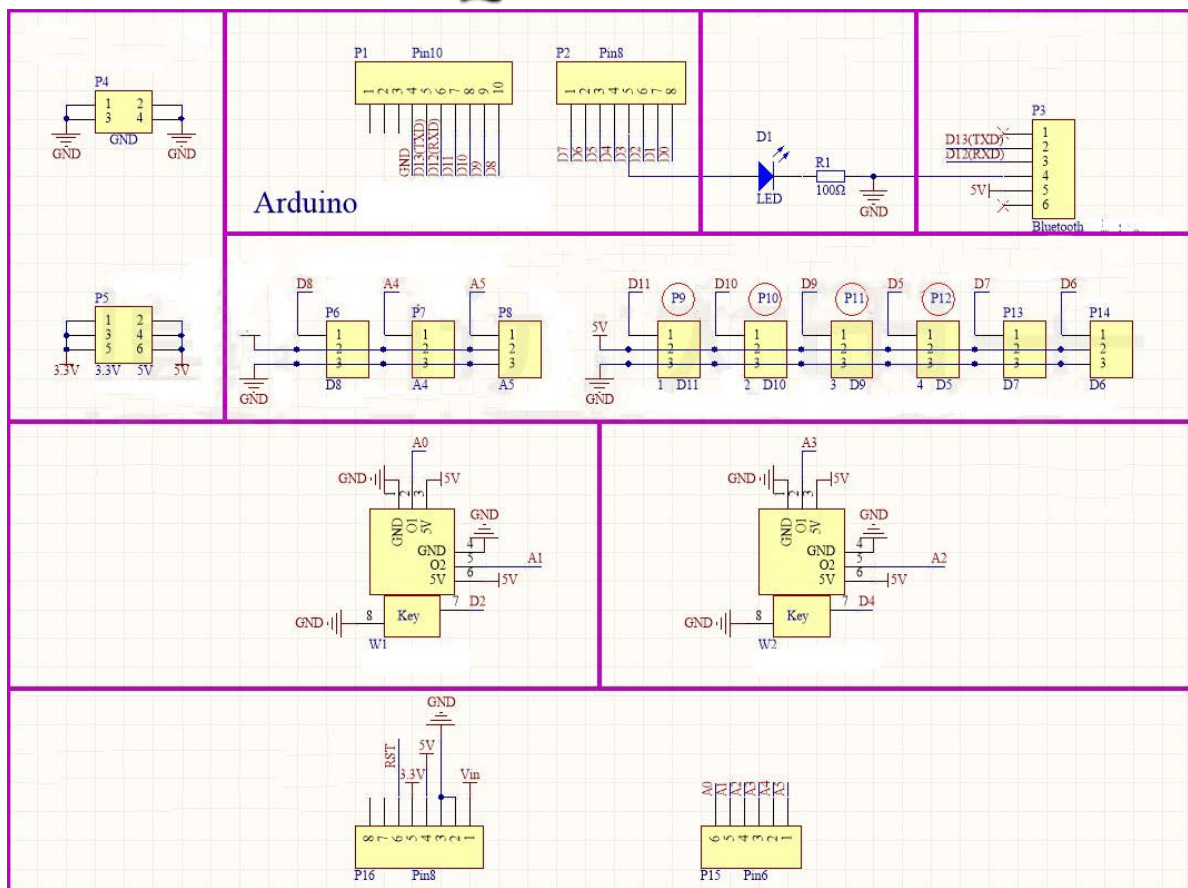
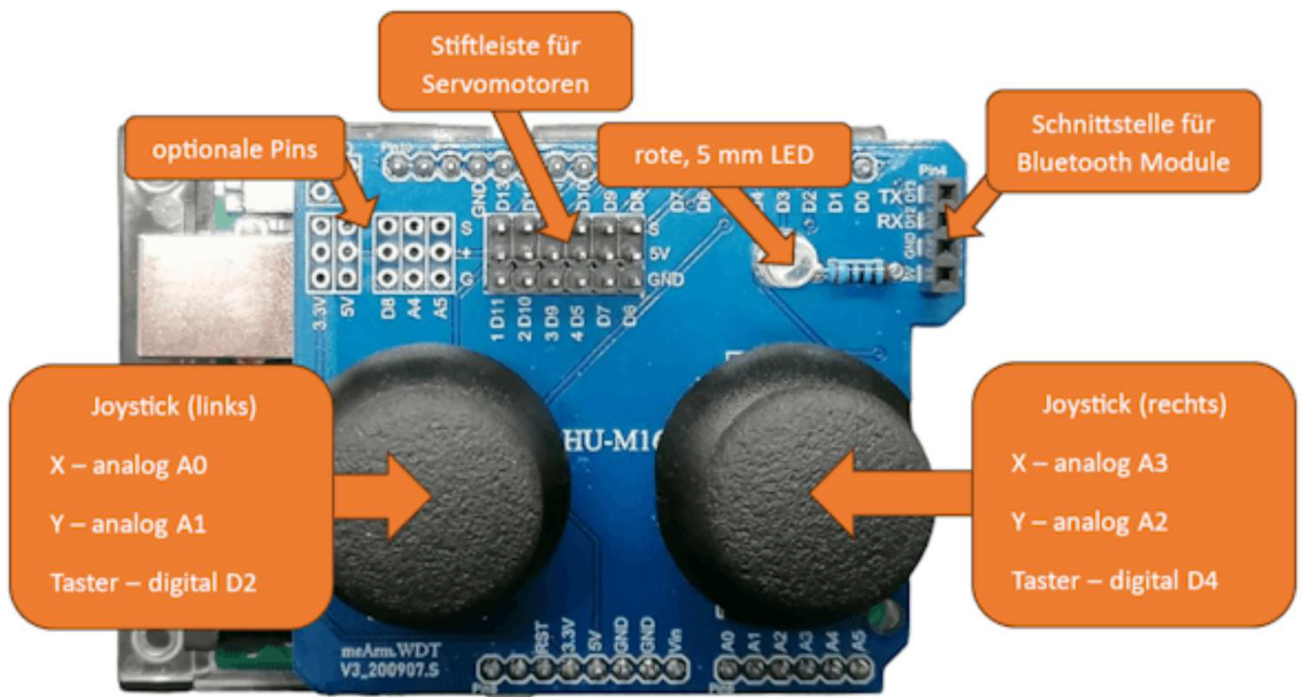
A complete 5-servo robotic arm control system with three control methods:

- Joystick Control - Dual joysticks for real-time control
- Serial Commands - Arduino IDE Serial Monitor
- Python GUI - Computer interface with visual controls

## 2. Hardware

- Arduino UNO R3 (main controller)
- 2 dual-axis joystick modules
- 5 MG995 servo motors
- Mechanical arm components

## 3. Pin Definition



## 4. Control Logic

### Control Mapping Table:

| Control        | Pin | Servo   | Range     | Function     |
|----------------|-----|---------|-----------|--------------|
| Joystick 1 X   | A0  | Servo 0 | 0-180 deg | X-axis       |
| Joystick 1 Y   | A1  | Servo 1 | 0-180 deg | Y-axis       |
| Joystick 1 Btn | D2  | Servo 4 | 0-90 deg  | +2 deg/press |
| Joystick 2 X   | A3  | Servo 2 | 0-180 deg | X-axis       |
| Joystick 2 Y   | A2  | Servo 3 | 0-180 deg | Y-axis       |
| Joystick 2 Btn | D4  | Servo 4 | 0-90 deg  | -2 deg/press |

## 5. Arduino Setup

1. Open servo\_control.ino in Arduino IDE
2. Select Board: Arduino UNO
3. Select COM port
4. Upload sketch
5. Open Serial Monitor (9600 baud)

## 6. Serial Commands

0 90 - Set Servo 0 to 90 deg  
1 45 - Set Servo 1 to 45 deg  
2 90 - Set Servo 2 to 90 deg  
3 90 - Set Servo 3 to 90 deg  
4 45 - Set Servo 4 to 45 deg (max 90)  
90 - Set ALL to 90 deg  
s - Show status  
h - Show help

## 7. Python G

Requirements: Python 3.x, pyserial

Setup: pip install pyserial

Run: python servo\_control.py

### Power Supply:

After uploading code, switch from computer USB to external power supply (9V 2A or higher). MG995 servos require adequate power.

### Overload Protection:

Do not apply excessive force. Servo stall causes current surge and board restart.

Support:

GitHub: <https://github.com/490437477-sys/ARM3>

